

## Chapter # 6

Q#25

With replacement:-

1. Both ball will be white

White balls = 4

Red balls = 2

$P(\text{White}) = ?$

$P(A) = \frac{n(A)}{n(S)} = \frac{\text{Total no. of favourable outcomes}}{\text{T. No. of outcomes in sample space}}$

$$n(S) = 4 + 2 = 6$$

$$P(\text{White}) = \frac{4}{6} = \frac{2}{3}$$

$$P(\text{both white}) = P(\text{White}) \cdot P(\text{White})$$

$$= \frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9} \quad \text{Ans}$$

2. Both ball will Same Colours:-

$$P(\text{both red}) = P(\text{Red}) \cdot P(\text{Red})$$

$$= \frac{2}{6} \cdot \frac{2}{6} = \frac{1}{9}$$

$$P(\text{same colour}) = P(\text{both white}) + P(\text{B. Red})$$

$$= \frac{4}{9} + \frac{1}{9} = \frac{5}{9}$$

3. That Both are different Colours:-

o First draw the white, second draw red.

o First draw red, second draw white

$$P(\text{diff Colours}) = P(W) \cdot P(R) + P(R) \cdot P(W)$$

$$= \frac{2}{3} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{2}{3} \Rightarrow \frac{2}{9} + \frac{2}{9}$$

$$= \frac{4}{9}$$

**WITHOUT REPLACEMENT:**

1. Both balls white:-

$$P(\text{White}) = \frac{4}{6} = \frac{2}{3}$$

$\Rightarrow$  (ek ball niklanay Kay bad 5)

$$P(\text{2nd W. ball}) = \frac{3}{5}$$

$$P(\text{Both White}) = \frac{2}{3} \times \frac{3}{5} = \frac{6}{15} = \frac{2}{5}$$

2. Both are same colours:-

$$P(\text{White}) = \frac{4}{6} = \frac{2}{3}$$

$$P(\text{both white}) = \frac{2}{5}$$

$$P(\text{Red balls}) = \frac{2}{6} \cdot \frac{1}{5} = \frac{1}{15}$$

$$P(\text{same colours}) = \frac{2}{5} + \frac{1}{15} = \frac{7}{15}$$

3. Both are different. (one red, one white):-

$$P(\text{d. colours}) = 1 - P(\text{same colours}) \\ = 1 - \frac{7}{15} = \frac{8}{15}$$

Q No #26

(b) Total eggs = 12  
Good eggs = 7  
Rotten eggs = 5

(i) First 2 are goods and remaining 3 are rotten?

$$\text{Good eggs} = P_1 = \frac{7}{12}$$

$$P_2 = \frac{6}{11} \text{ (one good egg is removed)}$$

$$P_3 = \frac{5}{10} \text{ (rotten eggs)}$$

$$P_4 = \frac{4}{9} \text{ (second rotten)}$$

$$P_T = P_1 + P_2 + P_3 + P_4 \\ = \frac{7}{12} + \frac{6}{11} + \frac{5}{10} + \frac{4}{9}$$

Fifth egg rotten: -  
 $P_5 = \frac{3}{8}$

$$P_{T2} = \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3}{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8} = \frac{2520}{95040} = 0.2647$$

$P_{T2} = 0.2647$  Answer

(ii) Any two are good and remaining three are rotten: -

Select two good eggs out of 7.

$${}^7C_2 = \frac{7!}{2!(7-2)!} = \frac{7!}{2!(5!)}$$

$$= \frac{7 \cdot 6 \cdot 5!}{2! \cdot 5!} = \frac{7 \cdot 6}{2!} = 21$$

Total select eggs out of 12 is

$${}^{12}C_5 = \frac{12!}{5!(12-5)!} = \frac{12!}{5!(7!)}$$

$$= \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7!}{5! \cdot 7!} = \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}$$

$${}^{12}C_5 = \frac{95040}{120} = 792 \text{ Ans}$$

Q#27  
(a)

Total no. of Students = 100

S' with defectied eyes = 60 = E

S with defectied ears = 50 = F

1) Propability of defectied ears and defectiv eyes:

$$|E \cup F| = |E| + |F| - |E \cap F|$$

$$|E \cup F| = 100$$

$$100 = 60 + 50 - |E \cap F|$$

$$|E \cap F| = 60 + 50 - 100 = 10$$

2)

$$P(\text{D. eyes but not D ears}) = |E| - |E \cap F|$$
$$= 60 - 10 = 50$$

$$P(A) = \frac{50}{100} = 0.5$$

$$P(\text{D. eyes and De. Ears}) = \frac{|E \cap F|}{100}$$

$$P(B) = \frac{10}{100}$$

$$P(B) = 0.1$$

1. Probability that all five cards are red.

Red cards = 26 (13 hearts, 13 diamonds)

⇒ The probability of selecting a red card on each draw, without replacement, is calculated as:-

$$\Rightarrow P(1) = \frac{26}{52} \quad \because 52 = \text{Total cards}$$

$$\Rightarrow P(2) = \frac{25}{52} \quad \text{when one card - (minus)}$$

$$\Rightarrow P(3) = \frac{24}{50} \quad (\text{already 2 cards been drawn})$$

$$\Rightarrow P(\text{all red}) = \frac{26}{52} \times \frac{25}{51} \times \frac{24}{50} \times \frac{23}{49} \times \frac{22}{48}$$

$$P(\text{all red}) = \frac{26 \cdot 25 \cdot 24 \cdot 23 \cdot 22}{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48} = 0.0517$$

2. Probability that all five cards are diamonds:-

There are 13 diamonds:

$$P(1) = \frac{13}{52}, \quad P(2) = \frac{12}{51}$$

$$P(3) = \frac{11}{50}, \quad P(4) = \frac{10}{49}$$

$$P(5) = \frac{9}{48}$$

$$P(\text{all diamonds}) = \frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} \cdot \frac{10}{49} \cdot \frac{9}{48}$$

$$= 0.0032 \quad \text{Ans.}$$

Q NO: 28

(a) Tossing a fair coin twice

Event A = A head occurs on the first toss

Event B = The same face does not occur on both tosses.  $\{\bar{H}\bar{H}, H\bar{H}\}$

Two events are independent i.e. -

$$P(A \cap B) = P(A) \times P(B) \Rightarrow \{H, H, HT\}$$

$$P(A) = \frac{1}{2} \quad (\text{A happens first toss is a head})$$

$$P(B) = ? \quad (B \text{ is happens two toss})$$

(HH (both heads), (HT), (TH), (TT))

→ If Two toss so faces are (HT) or (TH), there are 2 outcomes out of 4 possible outcomes.

P

Q#28 (a) Fair coin tossed twice

Event A: A head occurs on the first toss.  $A = \{H \cdot H, HT\}$

Event B: The same face does not occur on both tosses.

$$B = \{HT, TH\}$$

$$S = \{HH, HT, TH, TT\}$$

$$P(\text{each outcomes}) = \frac{1}{4}$$

$$P(A) = \frac{2}{4}, (HH, HT) = A$$

$$P(A) = \frac{1}{2}$$

$$B = (HT, TH)$$

$$P(B) = \frac{2}{4}, \frac{1}{2}$$

$$A \cap B = HT$$

$$P(A \cap B) = \frac{1}{4}$$

$$P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

(b) Two balls drawn without replacement:

Urn balls = 1, 2, 3, 4

Event A: The first ball drawn has a 1 on it.

$$A = \{(1,2), (1,3), (1,4)\}$$

Event B: The second ball drawn has a 1 on it.

$$B = \{(2,1), (3,1), (4,1)\}$$

$$S = \{(1,2), (1,3), (1,4), (2,1), (3,1), (4,1)\}$$

$S =$  All possible outcomes

$$S = \{(1,2), (1,3), (1,4), (2,1), (2,3), (2,4), (3,1), (3,2), (3,3), (3,4), (4,1), (4,2), (4,3)\}$$

$$P(A) = \frac{3}{12}$$

$$A = \{(1,2), (1,3), (1,4)\}$$

$$n(E) = 3, \quad n(S) = 12$$

$$P(A) = \frac{3}{12} = \frac{1}{4}$$

$$P(B) = \frac{3}{12} = \frac{1}{4}$$

$$P(A \cap B) = 0$$

$$P(A) \cdot P(B) = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$$

- (a) A and B are independent  
(b) A and B are not independent

## (Question # 29)

A = The event that the first die shows a 1.

B = The event that the second die show a 6.

C = The event that the sum of two dice is 7.

$$|S| = 6 \times 6 = 36$$

$$A = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)\}$$

$$A = 6$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

$$P(B) = \frac{6}{36} = \frac{1}{6}$$

$$C \Rightarrow n(C) = 6, n(S) = 36$$

$$P(C) = \frac{6}{36} = \frac{1}{6}$$

$$1. P(A \cap B)$$

The first die is 1 and the second is 6. =  $\{1, 6\}$

$$P(A \cap B) = \frac{1}{36}$$

$$2. P(A \cap C):$$

(A ∩ C) The first die is 1 and the sum is 7. = (1, 6)

$$P(A \cap C) = \frac{1}{36}$$

$$3. P(B \cap C) = \frac{1}{36}$$

4.  $P(A \cap B \cap C)$  The first die is 1 the second die is 6 and their sum is 7. (1, 6)

$$P(A \cap B \cap C) = \frac{1}{36}$$

$$P(A) \cdot P(B) \cdot P(C) = \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{216}$$

$P(A \cap B \cap C) \neq P(A) \cdot P(B) \cdot P(C)$  the events are not independent.

Q#31

(a) The probability of getting a head from a coin toss is  $\frac{1}{2}$   
The probability of number less than 5 (1, 2, 3, 4) from a die roll is  $\frac{4}{6} = \frac{2}{3}$

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{2}{3}$$

$$P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$$

1b)

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{3}$$

$$P(C) = \frac{1}{4}$$

$$P(A' \cap B' \cap C') = P(A') \cdot P(B') \cdot P(C')$$

$$= \left(1 - \frac{1}{2}\right) \cdot \left(1 - \frac{1}{3}\right) \cdot \left(1 - \frac{1}{4}\right)$$

$$= \left(\frac{1}{2}\right) \times \left(\frac{2}{3}\right) \times \left(\frac{3}{4}\right) = \frac{6}{24} = \frac{1}{4}$$

The probability that at least one student solves the problem is

$$1 - P(A' \cap B' \cap C') = 1 - \frac{1}{4} = \frac{3}{4}$$

## Question # 32

1a)

4 Kings in a deck of 52 cards.

(1) First Card Replaced.

$$\text{Kings} = 4$$

$$\text{first King picking} = \frac{4}{52}$$

$$P(K) = \frac{4}{52}$$

52

After replacement  $P(Q) = \frac{4}{52}$

52

(4 Queen in a deck)

$$P(K) \cdot P(Q) = \frac{4}{52} \cdot \frac{4}{52} \Rightarrow \frac{16}{2704}$$

We could pick a queen first then a King

$$= \frac{2 \times 16}{2704} = \frac{32}{2704}$$

$$= \frac{1}{85}$$

(ii) Not Replaced:-

$$P(K) = \frac{4}{52}$$

(after picking a King there are 51 cards left and 4 Queen)

$$P(Q) = \frac{4}{51}$$

$$P(K) \cdot P(Q) = \frac{4}{52} \times \frac{4}{51} = \frac{16}{2652}$$

$$\text{Total probability} = \frac{16}{2652} + \frac{16}{2652}$$

$$P(\text{King then queen}) = P(Q) + P(K)$$

$$P(K) \cdot P(Q) = \frac{16}{2652}$$

$$P(\text{Queen then King}) = P(Q) \cdot P(K)$$

$$= \frac{16}{2652}$$

$$P(\text{total}) = \frac{16}{2652} + \frac{16}{2652} = \frac{32}{2652}$$

$$= \frac{8}{663}$$

(b)

Total num of people = 10 + 8 + 9 + 6

$$T = 33$$

$${}^{33}C_6 = \frac{33!}{6!(33-6)!} = \frac{33!}{6!(27)!}$$

$$= \frac{33!}{6! \cdot 27!} = \frac{33 \cdot 32 \cdot 31 \cdot 30 \cdot 29 \cdot 28 \cdot 27!}{6! \cdot 27!}$$

$$= \frac{33 \cdot 32 \cdot 31 \cdot 30 \cdot 29 \cdot 28}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 1,107,568$$

(i)

2 = professor from 10

$${}^{10}C_2 = \frac{10!}{2!(10-2)!} = \frac{10 \cdot 9 \cdot 8!}{2! \cdot 8!}$$

$$= \frac{10 \cdot 9}{2 \cdot 1} = \frac{90}{2} = 45$$

2 = businessman from 8  
 ${}^8C_2 = 28$

1 = Doctor from 6  
 ${}^6C_1 = 6$

1 = lawyer from 9  
 ${}^9C_1 = 9$

$$= 45 \times 25 \times 6 \times 9$$

$$= 68040$$

So, = Num of ways to specific community  
Total num of ways.

$$= \frac{68040}{1.107568} = 0.0614$$

$$1.107568$$

(ii) No professor.

Committee of 6 from the non-professors =  $8+9+6 = 23$

$${}^{23}C_6 = 100947$$

= No. of ways to select of committee

Total num select any committee

$$= \frac{100947}{1.107568} = 0.0911$$

$$1.107568$$

(iii) At least one Professor.

$P = 1 - \text{no professors}$

$$= 1 - 0.0911$$

$$= 0.9089$$

A

Q#35

Total cards = 80

1 to 80 between

1, 4, 9, 16, 25, 36, 49, 64

There are 8 perfect squares.

Probability =  $\frac{\text{num of perfect square}}{\text{Total no of cards}}$

$$= \frac{8}{80} = \frac{1}{10}$$

= 0.1 Answer

## Question # 36

(a)

Three item from twenty

$$= {}^{20}C_3 = \frac{20!}{3! 17!}$$

$${}^{20}C_3 = 1140 = n(S)$$

If there is no faulty items

= faulty items - no faulty items

$$= 3 - 20 = 20 - 3$$

= 17 no faulty items.

$${}^{17}C_3 = \frac{17!}{3! 14!} = 686$$

if At least one faulty

= No. of ways - no-faulty items

$$= 1140 - 680$$

$$n(A) = 460$$

$$\text{probability} = \frac{460}{1140} = \frac{23}{57}$$

(b)

April = 30 days

1st F is not born same day as the host  $29/30$

2nd friend is not born on the same day as host is  $29/30$ .

probability of 5 friends are  
 $= \left(\frac{29}{30}\right)^5$

$$= 0.9836$$

Question # 37

**INDEPENDENT EVENTS:-**

These events are where the occurrence of one event doesn't affect the probability of the occurrence

of another events.

Mutually exclusive events are events where the occurrence of one event prevents the occurrence of another event. They can't happen at the same time.

### Question # 40

(C)

Total light bulbs = 15

Defective bulbs = 5

Non-defective bulbs =  $15 - 5 = 10$

No. of bulb chosen = 3

$${}^{15}C_3 = \frac{15!}{3!(15-3)!} = \frac{15!}{3! \cdot 12!}$$

$${}^{15}C_3 = \frac{15 \cdot 14 \cdot 13}{3 \cdot 2 \cdot 1} = 455 = n(s)$$

(a) None is Defective:

To have no defective bulb, all 3 bulbs must come from 10.

$${}^{10}C_3 = \frac{10!}{3!(10-3)!} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1}$$

$${}^{10}C_3 = 120 \Rightarrow n(A)$$

$$P(\text{none is defective}) = \frac{120}{455} = \frac{n(A)}{n(S)}$$

$$P(n.D) = \frac{24}{91}$$

(b) At least one is defective

$$P(\text{one.D}) = 1 - P(\text{non-D})$$

$$= 1 - \frac{24}{91} = \frac{1 - 24}{91}$$

$$= \frac{91 - 24}{91} = \frac{67}{91}$$

$$P(\text{one.De}) = \frac{67}{91} \quad \text{Answer.}$$

## Question # 41

(i) One is an Ace and the other a Jack:-

Total no. of Cards = 52

No. of Aces = 4

No. of Jacks = 4

choose 1 Ace =  ${}^4C_1 = 4$

$${}^4C_1 = \frac{4!}{1!(4-1)!} = \frac{4 \cdot 3!}{1! \cdot (3!)} = 4$$

Choose 1 Jack =  ${}^4C_1 = 4$

Total outcomes =  $4 \times 4$

$$n(A) = 16$$

⇒ total num. of ways to draw 2 cards from the deck is

$${}^{52}C_2 = \frac{52!}{2!(52-2)!} = \frac{52!}{2! \cdot 50!}$$

$$= \frac{52 \cdot 51 \cdot 50!}{2! \cdot 50!} = 1326 = n(S)$$

$$P(\text{one Ace and one Jack}) = \frac{n(A)}{n(S)} = \frac{16}{1326}$$

$$= \frac{8}{663}$$

(ii) First is an Ace and the other is a Jack:

$$P(\text{First is an Ace}) = \frac{4}{52}$$

after drawing 1 ace 51 cards left including 4 Jacks.

$$P(\text{second card is Jack}) = \frac{4}{51}$$

$$P(1st \text{ Ace} \cdot 2nd \text{ Jack}) = \frac{4}{52} \times \frac{4}{51}$$

$$2652 = \frac{4}{663}$$

## Question # 42

(b)

Choosing three distinct integers from first 15 positive integers:  
1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15

$$\text{odd num} = 7$$

$$\text{even num} = 8$$

(i)

Sum is odd:

We can select 1 or 3.

⇒ choosing 1 odd num  
 ${}^7C_1 = 7$  ways

⇒ Choos 2 even num  
 ${}^8C_2 = 28$  ways

$$7 \times 28 = 196$$

Choosing 3 odd numbers:

$${}^7C_3 = 35 \text{ ways}$$

$$196 + 35 = 231$$

$${}^{15}C_3 = \frac{15 \cdot 14 \cdot 13}{3 \cdot 2 \cdot 1} = 455$$

→ Probability that sum is odd

$$\frac{{}^7C_3}{{}^7C_2} = \frac{35}{455}$$

(ii) PRODUCT IS ODD:

3 odd num from the 7 available odd num

$${}^7C_3 = \frac{7!}{3!(7-3)!} = 35$$

$$\frac{35}{455} = \frac{1}{13}$$

(iii) PRODUCT IS EVEN:

The product is even if

$$455 - 35 = 399$$

$$P(\text{Product even}) = \frac{399}{455}$$

(iv) 'THEIR SUM IS EVEN'

odd num 8 from choosing 2.

There are 7 and choose 1.

$${}^8C_2 \times {}^7C_1 = \frac{8 \times 7}{2 \times 1} = 7$$

$$= 28 \times 7 = 196$$

$$P(\text{sum is even}) = \frac{196}{455}$$

## Question # 42

### (C) WINNING THE GAME WHEN ROLLIN A DIE:-

You win if outcome is either even or divisible by 3.

outcomes are =  $\{1, 2, 3, 4, 5, 6\}$

⇒ Even Numbers =  $(3, 4, 6)$  3 outcomes

⇒ Divisible by 3 =  $(3, 6)$  2 outcomes

The num 6 is both even and divisible by 3, we not double count. 2, 3, 4 and 6

$$P(\text{Win}) = \frac{4}{6} = \frac{2}{3}$$

## Question # 41